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GMBA 5885 W03
Strategic Mgt Global Info Sys

Enhancing Decision Making with Data Analytics and Business Intelligence

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Kick-Off Discussion

- How do you usually make a decision?
  (Intuition? Data? Advice? Experience?)
- What kinds of decisions do you make?
  (Routine, personal, financial, strategic, etc.)
- Do you use different decision processes in different contexts?
  (Work vs. personal life? High vs. low stakes?)
- Examples
  - Choosing a restaurant
  - Selecting an investment
  - Allocating team resources
  - Setting prices or forecasting demand

Learning Objectives

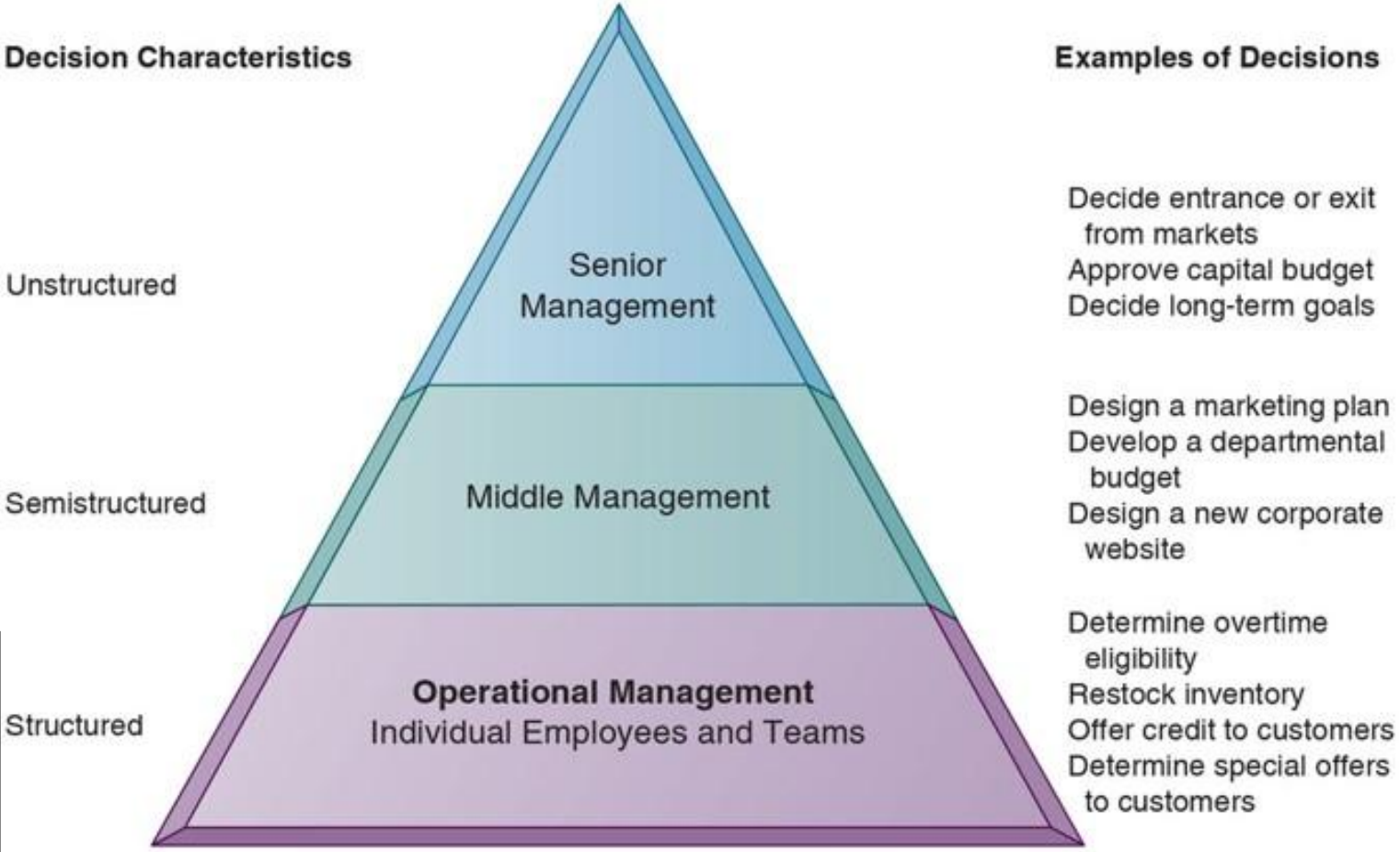
- Describe the different types of business decisions
- Describe the decision-making process
- Describe the different roles managers play
- Understand the value of improved decision making
- Understand why information systems do not always result in better decisions
- Discuss how data analytics, BI and AI support decision making
- Describe the capabilities of BI systems
- Understand the different types of BI users

Types of Business Decisions

Decision maker must provide judgment, evaluation, and insight to solve problem

Only part of problem has clear-cut answer provided by accepted procedure

Repetitive and routine; involve definite procedure for handling so they do not have to be treated each time as new



Quiz

Which of the following is an example of a **semistructured** decision?

A. Determining overtime eligibility

B. Deciding long-term goals

C. Approving a capital budget

D. Designing a marketing plan

Decision-Making Process

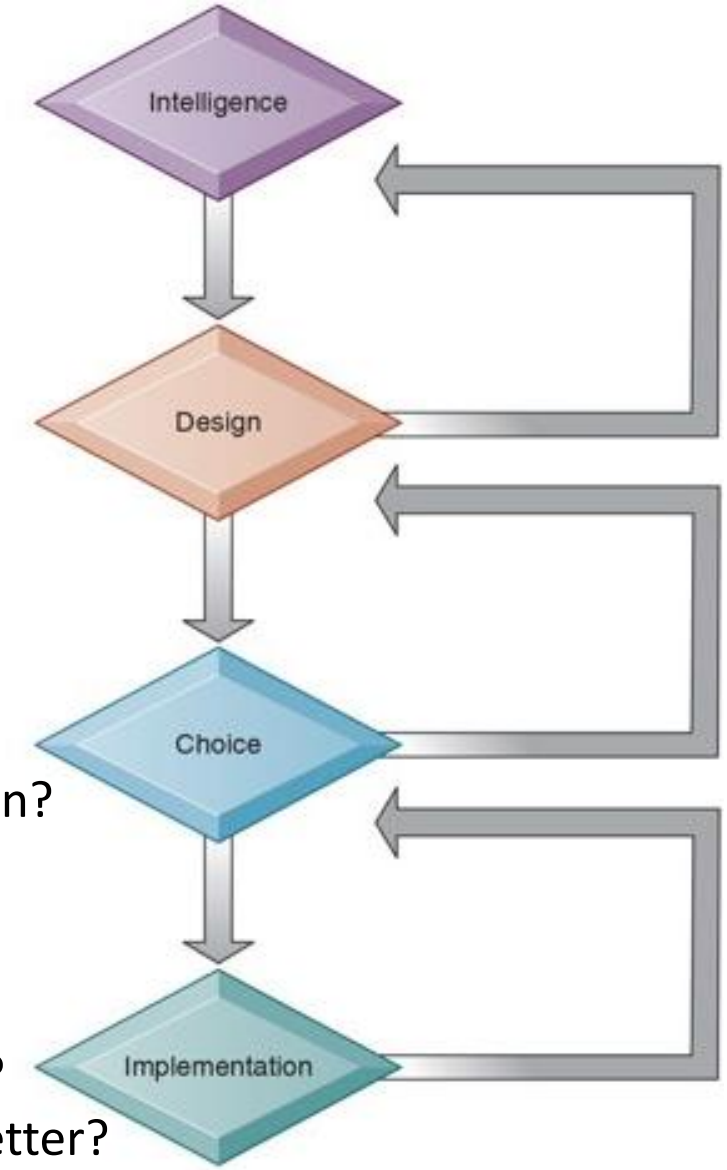
Use a simple example, such as “Which apartment should I rent?”, and map the actions involved in each of the four decision-making stages.

Problem discovery:
What is the problem?

Solution discovery:
What are the possible solutions?

Choosing solutions:
What is the best solution?

Solution testing:
Is the solution working?
Can we make it work better?



Quiz

Which phase of decision making involves finding or recognizing a problem?

A. Intelligence

B. Choice

C. Design

D. Implementation

Managerial Roles

- Classical Model (Henry Fayol)
  - Key functions: Planning, Organizing, Coordinating, Deciding, Controlling
  - Formal, structured view of management work
- Behavioral Model
  - Fast-paced: hundreds of activities per day
  - Fragmented tasks: most < 9 minutes
  - Preference for current, specific info
  - Oral communication > written
  - Extensive informal networks for information & influence
- Actual managerial behavior is **informal**, **reactive**, **dynamic**, not fully captured by classical functions.

Managerial Roles (Mintzberg 1971)

Role	Behavior	Support Systems
Interpersonal Roles	Interpersonal	
Figurehead		Telepresence systems
Leader		Telepresence, social networks
Liaison		Smartphones, social networks
Informational Roles	Information processing	
Nerve center		Management information systems, executive support systems
Disseminator		Texting, email, social networks
Spokesperson		Webinars, telepresence
Decisional Roles	Decision making	
Entrepreneur		None exist
Disturbance handler		None exist
Resource allocator		Business intelligence, decision-support system
Negotiator		None exist

Quiz

All of the following managerial activities can be characterized as falling with the interpersonal role of managers except:

A. initiating new activities

B. acting as a leader

C. acting as a figurehead

D. acting as a liaison

Value of Improved Decision Making

Example Decision	Decision Maker	Number of Annual Decisions	Estimated Value of a Single Improved Decision	Annual Value
Allocate support to most valuable customers	Accounts manager	12	$100,000	$1,200,000
Predict call center daily demand	Call center management	4	$150,000	$600,000
Decide parts inventory levels daily	Inventory manager	365	$5,000	$1,825,000
Identify competitive bids from major suppliers	Senior management	1	$2,000,000	$2,000,000
Schedule production to fill orders	Manufacturing manager	150	$10,000	$1,500,000
Allocate labor to complete a job	Production floor manager	100	$4,000	$400,000

Why IS Don’t Always Improve Decisions

- Information quality
  - High-quality decisions require high-quality information

Quality Dimension	Description
Accuracy	Do the data represent reality?
Integrity	Are the structure of data and relationships among entities and attributes consistent?
Consistency	Are data elements consistently defined?
Completeness	Are all the necessary data present?
Validity	Do data values fall within defined ranges?
Timeliness	Are data available when needed?
Accessibility	Are the data accessible, comprehensible, and usable?

Why IS Don’t Always Improve Decisions

- Information quality
  - High-quality decisions require high-quality information
- Management filters
  - Managers have selective attention and have variety of biases that reject information that does not conform to prior conceptions
- Organizational inertia and politics
  - Strong forces within organizations resist making decisions calling for major change

High-Velocity Automated Decision Making

- Made possible through computer algorithms precisely defining steps for a highly structured decision
  - Humans taken out of decision
- For example: High-speed computer trading programs
  - Trades executed in nanoseconds
- Require safeguards to ensure proper operation and regulation

Risks of Using AI in Decision Making

AI-driven decisions introduce several challenges

- Bias embedded in data or models
- Data quality concerns
- Accuracy and reliability issues
- Lack of transparency (“black-box” decisions)
- Ethical concerns and fairness implications

Quiz

Which of the following is not a main reason why information systems do not always result in better decisions?

A. Information quality

B. Organizational culture

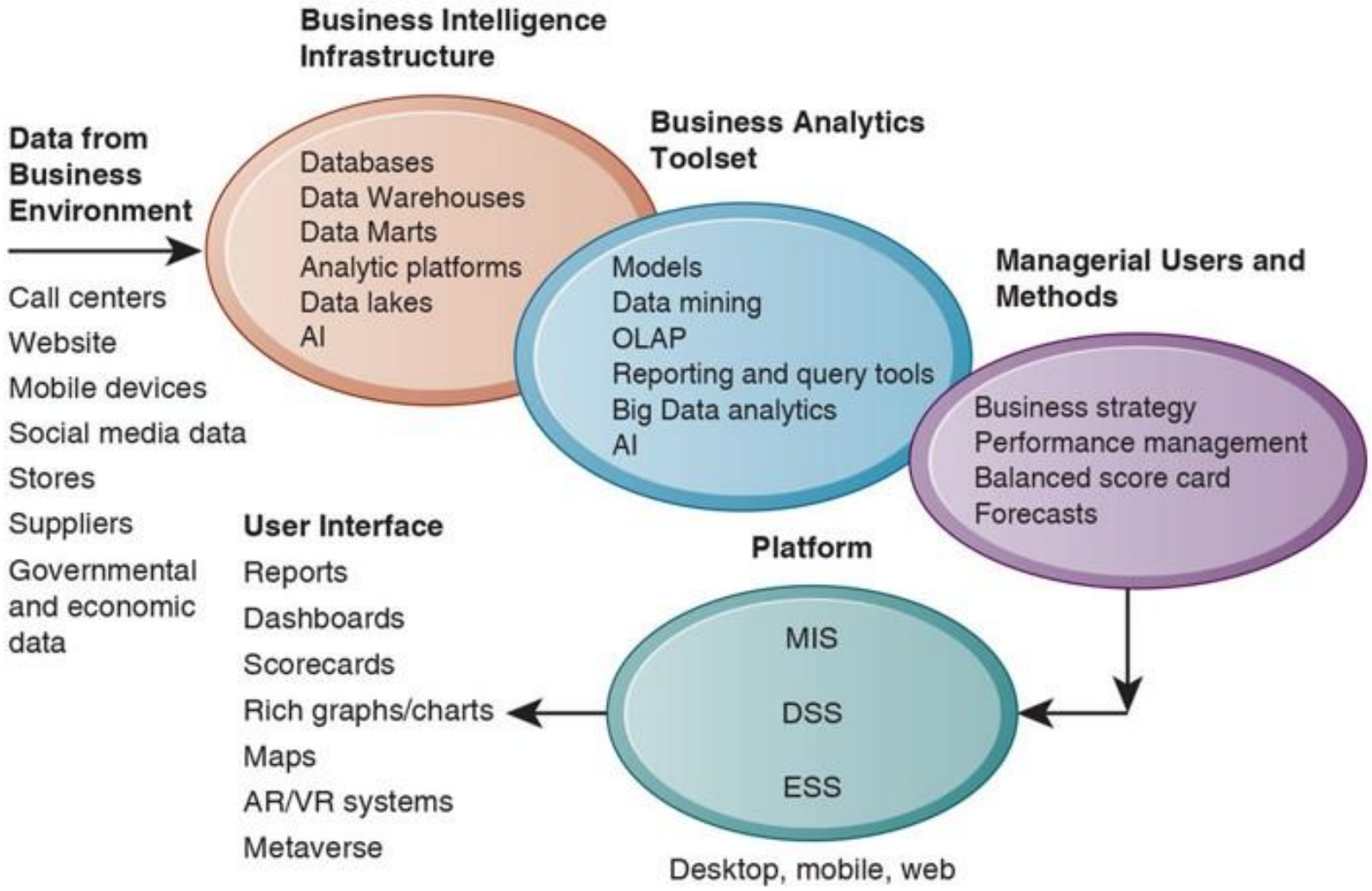
C. External forces

D. Management filters

How DA, BI, AI Support Decision Making

- Data Analytics (DA) – data preparation + basic analysis
  - Foundation layer
  - Collects, cleans, stores, processes data
- Business Intelligence (BI) – insight delivery, what happened?
  - Uses DA outputs
  - Turns processed data into reports, dashboards, KPIs for managers
- Business Analytics (BA) – why & what next?
  - Builds on DA + BI data
  - Focus: descriptive (why it happened), predictive (what will happen), prescriptive (how to make it happen)

How DA, BI, AI Support Decision Making



How DA, BI, AI Support Decision Making

- Tableau
  - Leading BI and data visualization tool
- Virtual Reality (VR)
  - 3D, immersive data visualization
  - Interactive graphics software and hardware
  - Creates realistic, simulated environments
- Augmented reality (AR)
  - Overlays digital data onto the real world
  - Enhances physical environments with visual information

Capabilities of BI Systems

Six Analytic Functionalities of BI Systems

- Production reports – Predefined, routine reports
- Parameterized reports – User-specified filters (e.g., region, time)
- Dashboards / scorecards – Visual performance snapshots
- Ad hoc queries & reports – User-created queries and outputs
- Drill-down – From summary to detailed data
- Forecasts & scenarios – Forecasting, what-if analysis, modeling, recommendations

Capabilities of BI Systems

- Predictive Analytics
  - Uses statistical models to forecast future outcomes
- Prescriptive Analytics
  - Recommends actions to achieve desired results
- Big Data Analytics
  - Large, diverse datasets from digital and physical sources
- Operational Intelligence
  - Real-time analytics from continuous data streams
- Location Analytics
  - Insights derived from geographic/spatial data

Different BI Users

Power Users: Producers (20% of employees)	Capabilities	Casual Users: Consumers (80% of employees)
IT developers	Production Reports	Customers/suppliers Operational employees
Super users	Parameterized Reports	
Business analysts	Dashboards/Scorecards	Senior managers
Analytical modelers	Ad hoc queries; Drill down Search/OLAP	Managers/Staff
	Forecasts; What if Analysis; statistical models	Business analysis

Decision Support for Operational & Middle Management

- Focus on monitoring performance (machine downtime, hourly sales, website traffic)
- Decisions are routine and structured
- MIS provides reports for tracking performance
- Managers can query data interactively to understand causes
- Use exception reports to flag unusual conditions (e.g., low sales, overspending)
- Supports day-to-day control and timely corrective actions

Support for Semistructured Decisions

- “Super users” create custom reports and explore data with advanced analytics
- Decision Support Systems (DSS) support semistructured decisions
- Rely on models: what-if analysis, scenario modeling, goal seeking
- Sensitivity analysis: test changes in variables to see outcome ranges
- Multidimensional analysis / OLAP / pivot tables reveal patterns across dimensions
- Modern BI systems integrate these capabilities for deeper insights and targeted actions

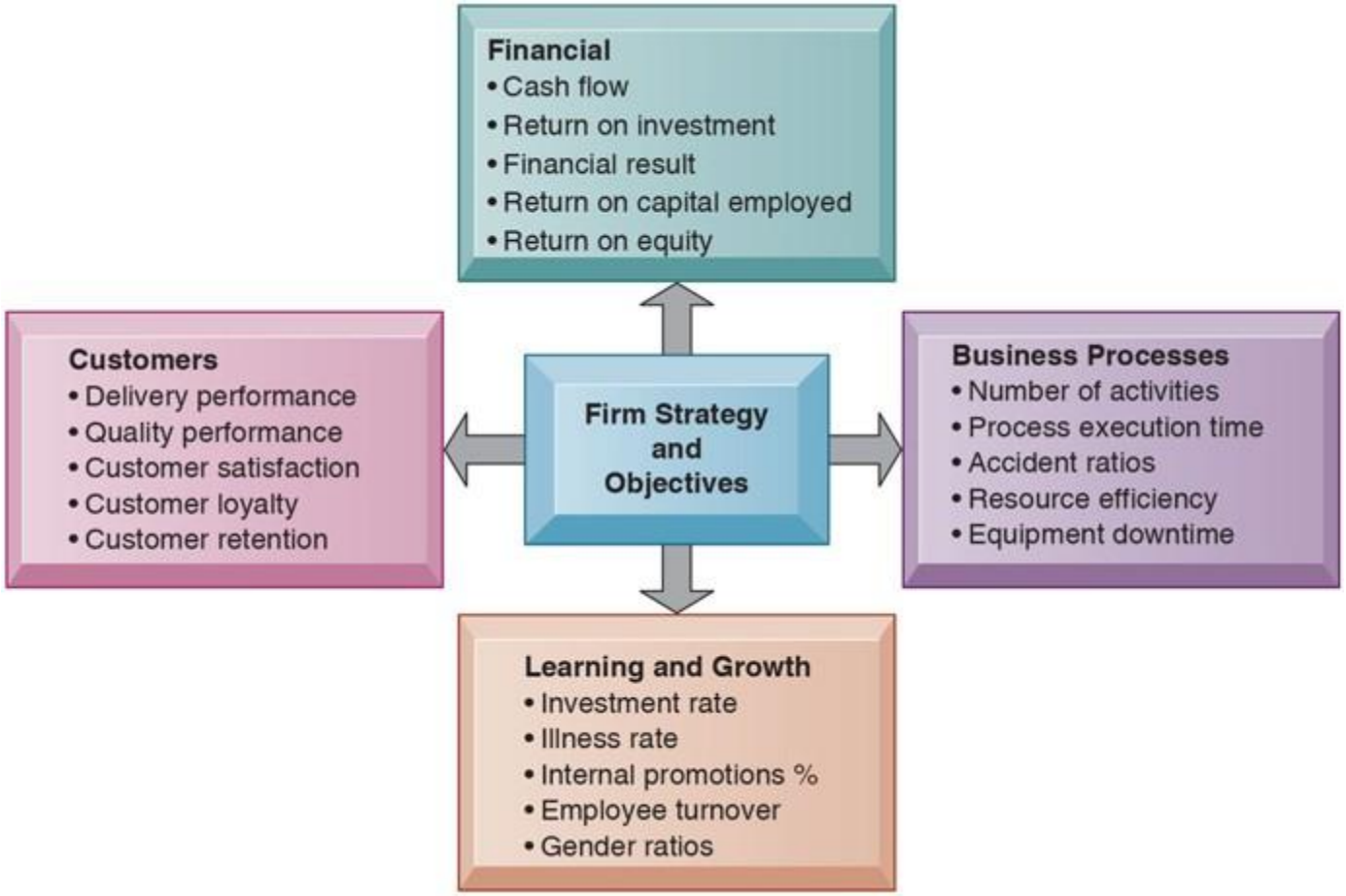
Decision Support for Senior Management

Executive Support Systems (ESS)

- Provide critical, high-level performance information
- Integrate data from ERP, SCM, CRM + external sources
- Dashboards, KPIs, and drill-down views
- Support strategic monitoring and decision making

Decision Support for Senior Management

Balanced Scorecard Framework



Decision Support for Senior Management

Business Performance Management (BPM)

- Translates strategy → operational targets → KPIs
- Monitors progress using enterprise data
- Similar to BSC but more strategy-driven
- Helps identify gaps, trends, and improvement opportunities

Summary

- Types of Decisions: structured, semistructured, unstructured
- Decision Process: intelligence → design → choice → implementation
- Managerial Roles: interpersonal, informational, decisional
- Value of Better Decisions: improved performance and reduced costs
- Limits of IS: information quality, management filters, organization inertia and politics
- DA, BI & AI: provide data, insights, models, and automation for decisions
- BI Capabilities: reporting, dashboards, OLAP, drill-down, forecasting, scenarios
- BI Users: operational users, analysts/super users, executives